



Complex Data Analysis

Grupo 12

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Introduction

Social networks in the real world contain a lot of information encoded within their interactions. By studying this data, it will be possible to better understand several questions, such as how communities form or how information disseminates through a network. This is the motivation behind the analysis of this dataset.

Dataset Description

From the Stanford Network Analysis Project (SNAP), we've chosen sx-mathoverflow as our dataset.

MathOverflow is a question-and-answer (Q&A) platform targeted at professional mathematicians and researchers, where users post research-level problems and exchange knowledge.

The SNAP dataset captures interactions between users extracted from the public Stack Exchange data dumps.

Dataset Description

The full dataset represents MathOverflow as a directed temporal network.

Each node corresponds to a user account, and each temporal edge is a triplet (u, v, t) indicating that user u interacted with user v at timestamp t .

Interactions include:

- answering another user's question.
- commenting on a question.
- commenting on an answer.

The complete temporal dataset contains 24,818 nodes and 506,550 time-stamped interactions in a period of 2,350 days. A static projection of all interactions produces a directed graph with 239,978 edges.

What We Expected (Literature)

- Heavy-tailed degree distribution
 - Few expert users contribute most answers.
- Sparse but clustered network
 - Low density with localized topical cohesion.
- Negative assortativity
 - Experts mainly interact with less active users.
- Low reciprocity
 - Interactions are mostly unidirectional.
- Moderate modularity & small-world structure
 - Thematic communities with short path lengths.

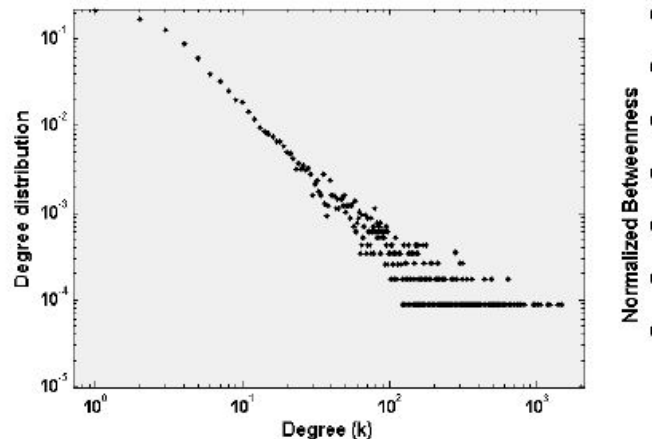


Figure 1. Heavy-tailed degree distribution in the MathOverflow network (Montoya et al., 2013).

Source: Montoya et al. (2013), Tausczik & Pennebaker (2012)

Subset Creation and Justification

The original Math Overflow interaction dataset is extensive, and so to focus the analysis on the most representative dynamics of the community a two-step procedure was implemented to create a representative subset.

The adopted methodology combines:

- Temporal Filtering - the initial subset was delimited to the period from January 1, 2010, to December 31, 2012, which is the period of more activity.
- User Activity Filtering - we only considered interactions where both users were “active users”, defined as someone which participated in at least 25 interactions throughout the time period above.

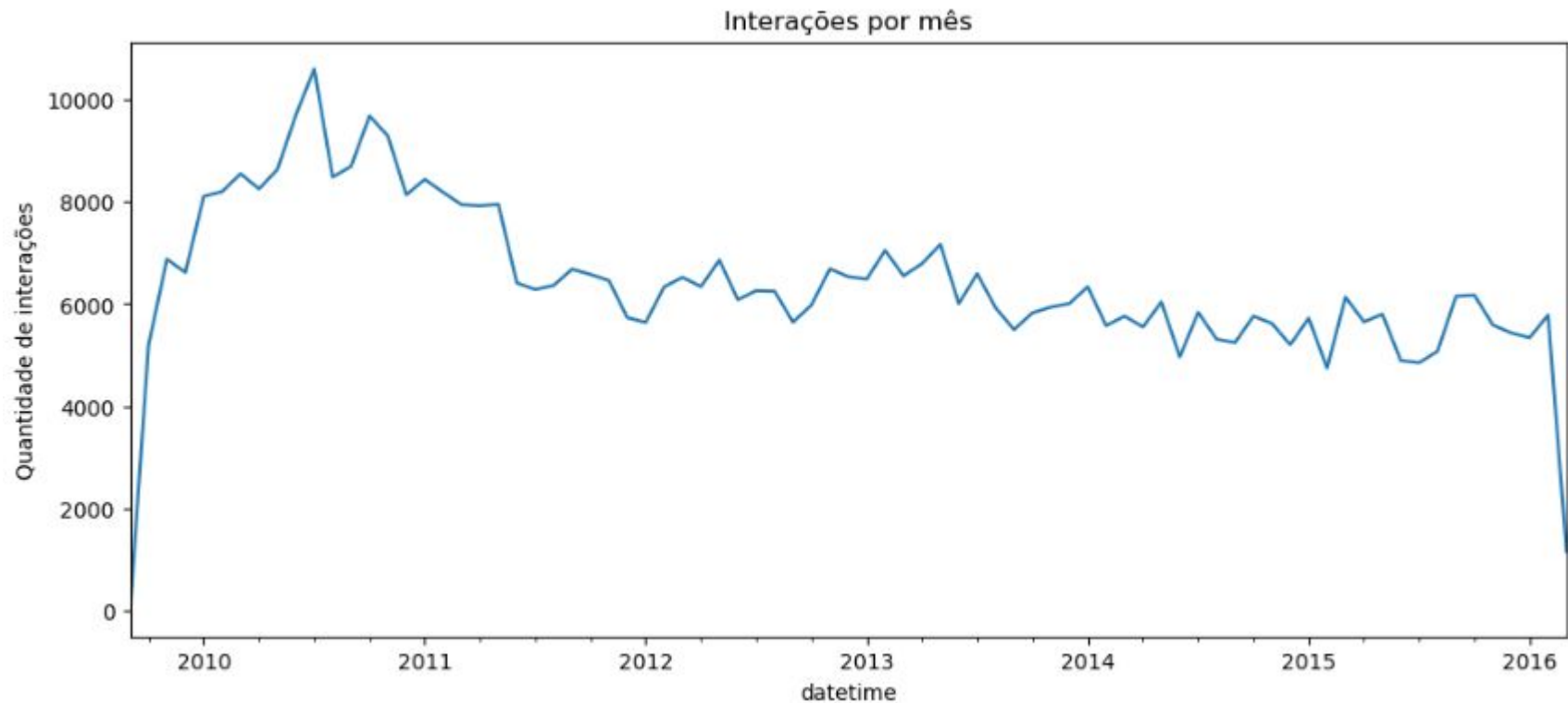


Figure 2. Temporal evolution of the volume of interactions (edges) in the Math Overflow dataset. The graph clearly shows a spike in growth between 2010 and 2011.

Subset Analysis

Subset Analysis

The fundamental statistics quantify the basic structure of the active community:

- Clustering Coefficient (C): 0.0970
- Density (ρ): 0.0146

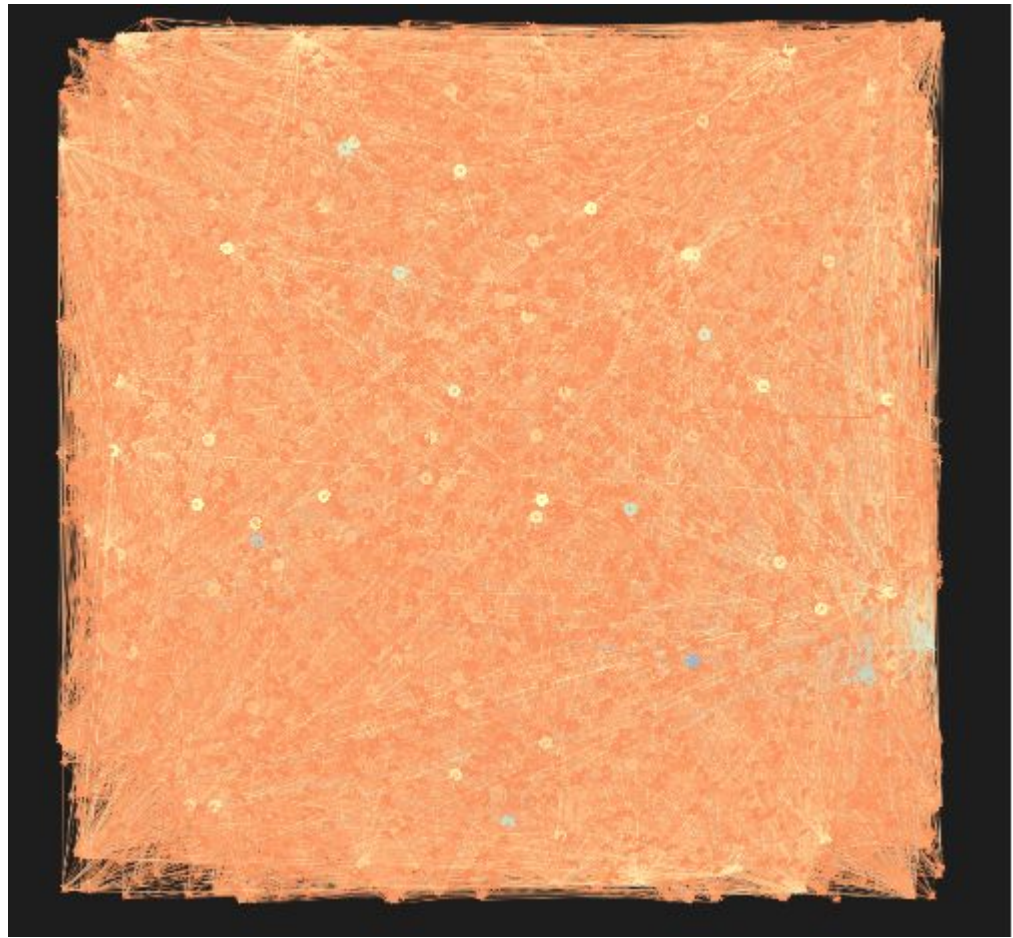
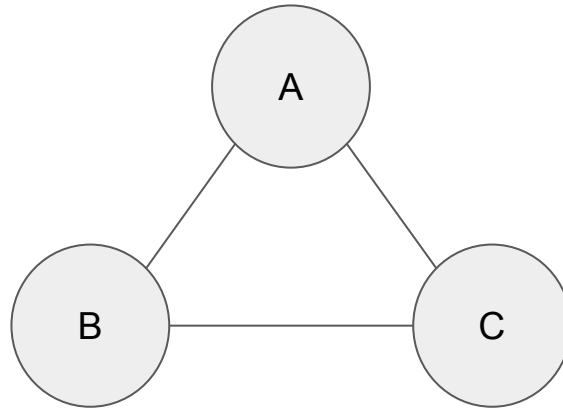


Figure 3. Visualization of the chosen subset.

Clustering Coefficient

The network exhibits a Clustering Coefficient of 0.0970. This non-random value signifies a higher-than-expected tendency for the formation of closed triangles within the active subset. This high local clustering indicates the formation of local communities where users often interact around shared topics or experts, demonstrating high proximity among interconnected members.



Average Path Length (APL) and Connectivity

When considering the undirected projection of the graph, the APL is 2.25.

This reveals that the active community is a Small-World network, where any user can reach any other user through an average of only 2.25 steps (ignoring direction).

This demonstrates high global efficiency in information access.

Despite this efficiency, the directed APL calculation failed because the subset, while being the Largest Weakly Connected Component (LWCC), is not a Strongly Connected Component (SCC). The existence of a path $A \rightarrow B$ (User A answers B's question) does not guarantee the existence of a path $B \rightarrow A$.

Reciprocity

The Reciprocity of the network is 0.3883.

Reciprocity measures the probability that a directed edge is part of a reciprocal pair ($A \rightleftharpoons B$). So this value leads us to two conclusions:

- This value confirms that the majority of relationships (~ 61%) remain unidirectional, supporting the initial hypothesis of a "knowledge provider $\rightarrow$ seeker" or "Expert $\rightarrow$ Novice" dynamic.
- However, the 38.83% reciprocity also indicates that a substantial portion of the relations among the core contributors are mutual.

Discussion on Assortativity

The analysis of assortativity for the different degree combinations reveals a dominant characteristic of dissortativity (negative correlation) across all combinations:

- Out-In Assortativity (*r_{out-in}*): -0.1161
- Out-Out Assortativity (*r_{out-out}*): -0.1106
- In-In Assortativity (*r_{in-in}*): -0.0896

The negative value of *r_{out-in}* establishes the network's hierarchy: the greatest knowledge providers (high *k_{out}*) link to the greatest solicitors (high *k_{in}*) and to peripheral users.

Even more revealing is the fact that *rout-out* and *rin-in* are also negative. In social networks, it is typically expected for hubs to connect to hubs (assortativity). In MathOverflow, this generalized disassortativity confirms that the dominant dynamic is the dispersion of knowledge rather than collaboration between peers of equal status.

Specifically:

- Out-Out Assortativity (-0.1106): The most prolific answerers (high *k_{out}*) do not tend to answer questions posed by other prolific answerers. This suggests that interactions are more about the transfer of knowledge to the base than about high-level technical debates among experts.
- In-In Assortativity (-0.0896): The biggest information solicitors (high *k_{in}*) do not tend to ask questions of other solicitors. This metric has a less extreme value but points to a structure where questions are directed by thematic necessity, and not by an affinity of "seeker" status.

Central Conclusion: The combination of low reciprocity and generalized dissortativity is the clearest evidence that the primary function of the MathOverflow subset is the efficient dispersion of knowledge from centers of authority to the user base.

Degree Distribution

To characterize user activity and interaction imbalance, we analysed the In and Out degree of the graph:

- Average In-Degree ($\langle k_{in} \rangle$): 35.94
- Average Out-Degree ($\langle k_{out} \rangle$): 35.94
- Maximum In-Degree ($\max k_{in}$): 498
- Maximum Out-Degree ($\max k_{out}$): 792

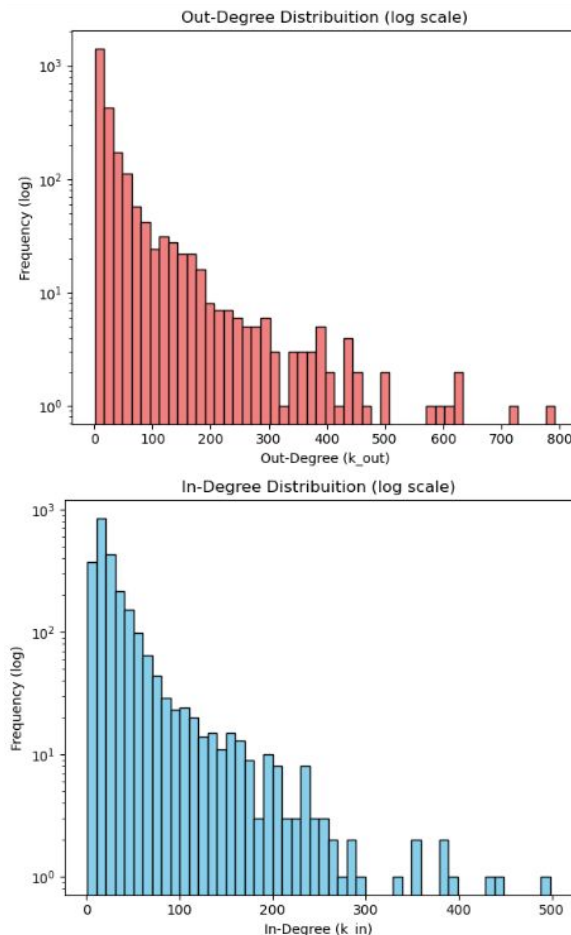


Figure 4. In and Out degree distribution from the created subset.

Centrality Analysis

To identify the most structurally critical users in mediating knowledge flow, we computed the Betweenness.

- Betweenness Centrality

User_ID	Betweenness Centrality
290	0.0515
11142	0.0480
297	0.0353
1409	0.0343
6094	0.0310

Figure 5. Top 5 users with highest betweenness centrality.

Community Detection

To identify the underlying structure and grouping within the core user community, we used the Louvain Algorithm. Here are the results we obtained:

- Global Modularity (Q): 0.2451
- Total Number of Communities: 7
- Size of the two Largest Communities: 549 and 509 nodes

Community Detection

The resulting modularity value of $Q = 0.2451$ suggests a **moderate but observable community structure**. While this value is not indicative of an extremely fragmented network, it still confirms that users interact more frequently within certain groups than across the entire network.

The algorithm successfully identified 7 distinct communities. The structure is characterized by two dominant, dense groups (549 and 509 nodes), with the remaining nodes distributed among five smaller clusters.

Figure 6. Representation of the communities found



Temporal Activity Patterns

Leveraging the temporal feature of the dataset, we analyzed the hourly volume of interactions (edges) to understand user engagement patterns over time.

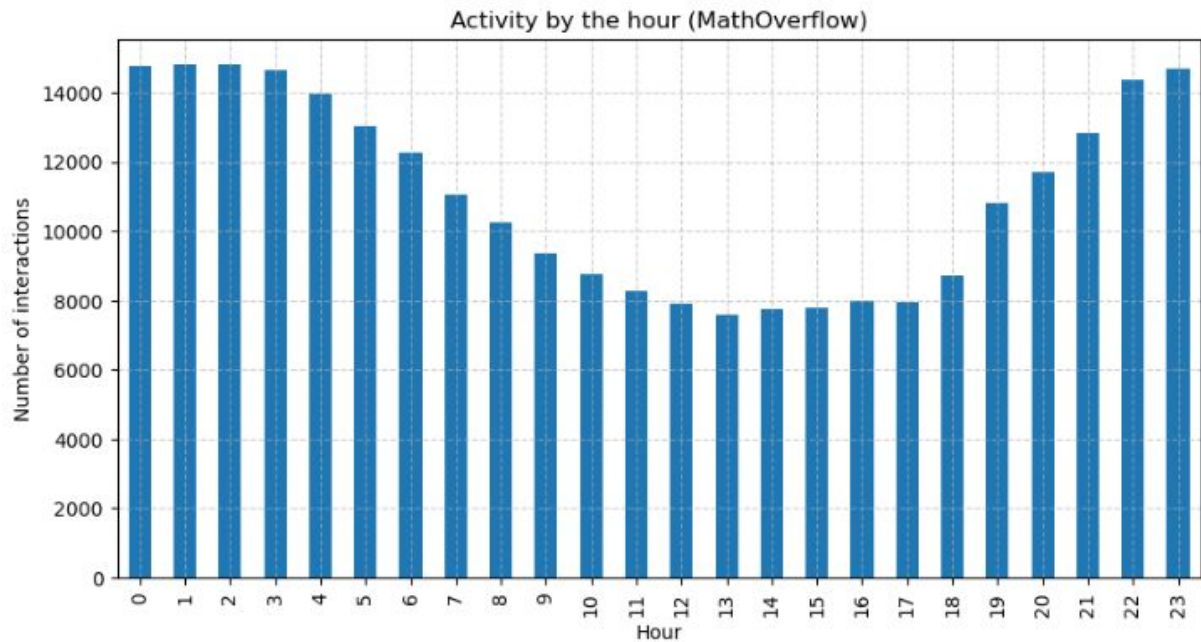


Figure 7. Hourly interaction volume (UTC).

Temporal Activity Patterns

This inverted cycle, where activity peaks during what corresponds to the nighttime and reaches its minimum during standard business hours, is atypical for a professional, research-oriented platform.

A plausible explanation is that a substantial portion of the platform's most active core users are located in Asian time zones. The global peak (01:00–02:00 UTC) aligns with the start of the workday in India (06:00–07:00) and mid-morning in East Asia (09:00–10:00). Alternatively, the high activity could be attributed to users in the **USA**, where MathOverflow has a large user base, who dedicate time to the site during their night.

Discussion & Conclusion

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The structural properties observed in the filtered MathOverflow subset closely match those reported in prior studies of the full platform. The network preserves all hallmark characteristics of MathOverflow: **heavy-tailed degree distributions, sparsity, moderate clustering, moderate modularity, small-world behavior, low reciprocity, and consistently negative assortativity**. These results confirm that participation is highly unequal and hierarchically organized, with a small group of experts responsible for most knowledge production and dissemination.

The similarity between our measurements and values reported in the literature indicates that the **temporal (2010–2012) and activity-based filtering does not distort the underlying network dynamics**. Degree distributions, clustering levels, assortativity patterns, and community structure all remain consistent with known MathOverflow behavior, supporting the **representativeness of the chosen subset**.